For ± 20% tolerance with multiplying factor of 1.46, the preferred values are

10, 15, 22, 33, 47 and 68.

Thus for 10% tolerance the commercially manufactured values will be as shown in Table-1.1.

Table-1.1 : Commercially manufactured resistors for 10% tolerance

Ohms and kilo ohms			Mega ohms	
1.1	10	100	1.0	10
1.2	12	120	1.2	12
1.5	15	150	1.5	15
1.8	18	180	1.8	18
2.2	22	220	2.2	22
2.7	27	270	2.7	
3.3	33	330	3.3	
3.9	39	390	3.9	
4.7	47	470	4.7	
5.6	56	560	5.6	
6.8	68	680	6.8	
8.2	82	820	8.2	

1.2.7 Standard Power Ratings :

Standard power ratings are 0.125, 0.25, 0.5, 1, 2, 3, 4, 5, 10, 20, 25, 30, 40, 50, 75, 100, 150, 200 W and higher. Power ratings above 2 W are invariably of wire wound types.

1.2.8 Resistors Colour Code :

In the above method of indicating resistance value by printing it is difficult to read when the resistor is used on PCB which is densely packed with the components. To overcome this difficulty colour code is used. There are two methods in this.

(a) Colour band method and

(b) Body end dot colour method.

The second method is not used at present. But it is given here to help if you come across an old equipment in which the resistors with this type of coding are used.

(a) Colour band method :

In this method four colour band are printed on one side of the resistor. Three bands are printed side by side and the fourth band is printed slightly away from the third band. Three bands from left indicate its value whereas the fourth band indicates the tolerance. First two bands indicate the significant figures while the third band indicates the decimal multiplier. Table 1.2 shows significant figure decimal multiplier and tolerance for different colours.

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